Verapay Technical Docs

Functional Requirements Document (FRD)

SYSTEM SPECIFICATION & ACCEPTANCE CRITERIA • VERSION 2.0

Verapay Platform
Verify, then release.

Verapay: Functional Requirements Document (FRD)

System Specification & Technical Requirements | Version 2.0

1. Document Control & Overview

Attribute	Specification
System Name	**Verapay Core Platform** ("*Verify, then release.*")
Document Type	Functional Requirements Document (FRD)
Version	2.0 (Hackathon MVP & Production Blueprint)
Audience	Frontend Engineers, Backend Engineers, QA Engineers, System Architects
Primary Rails	Sender Funding: PayPal Checkout • Payout Settlement: Paystack / Flutterwave NIP Rails

2. System Architecture & High-Level Flow

Verapay is an escrow orchestration and automated fraud-prevention layer positioned between international payment rails (PayPal) and local Nigerian banking infrastructure (NIP NUBAN transfers).

```
graph TD
    Sender([Diaspora Sender]) -->|1. Fund Payment| VP_Core[Verapay Core Escrow]
    VP_Core -->|2. Enqueue Verification| AI_Engine[AI & Registry Verification Engine]

    subgraph "Verification Pipeline"
        AI_Engine -->|Gate 1| CAC[CAC & NUBAN Registry Check]
        AI_Engine -->|Gate 2| OCR[Multimodal OCR & Quote Agent]
        AI_Engine -->|Gate 3| Ground[Camera Forensics & Delivery Tap]
    end

    AI_Engine -->|3. Risk Assessment| TrafficLight{Gate 4: Sender Approval}
    TrafficLight -->|GREEN: All Good| Approve[Sender Approves]
    TrafficLight -->|AMBER: Price High| Modify[Review Alternatives / Override]
    TrafficLight -->|RED: Fraud Caught| Reject[Sender Rejects]

    Approve -->|4. Release Escrow| Payout[Paystack/Flutterwave Naira Payout]
    Reject -->|5. Instant Refund| Refund[PayPal Refund API]
    Payout --> Provider([School / Hospital / Merchant])
    Refund --> Sender
```

3. User Roles & Permissions Matrix

Role	Description	Key Capabilities
`DIASPORA_SENDER`	Senders residing abroad (UK, US, CA) funding payments.	Fund escrow, view AI risk signals, approve/reject payouts, create family groups, manage project milestones.
`FAMILY_MEMBER`	Dependents or relatives in Nigeria.	View family transparent ledger, upload tuition/medical bills, tap "Item Received" for consumer goods.
`CONTRACTOR`	Site foremen, builders, or vendors in Nigeria.	Submit milestone completion claims, capture live in-app photos, execute 15-second surprise video challenge.

Role	Description	Key Capabilities
`FIELD_SURVEYOR`	Certified third-party professional inspector.	Perform on-site structural/land inspections, submit verified independent audit reports.
`VERAPAY_OPERATOR`	Human compliance and triage analyst.	Review Amber-flagged transactions for Verapay Plus users, mediate stalled project disputes.

---

4. Functional Requirements Specification

4.1. Authentication, KYC & Plan Management (FR-AUTH / FR-PLAN)

`FR-AUTH-01`: Multi-Region Identity & Onboarding

- **Description:** Senders register using email, Google OAuth, or PayPal login. International phone numbers (UK +44, US/CA +1) require SMS OTP verification.
- **Nigerian Recipients/Contractors:** Register via phone number (+234) with SMS OTP. Optional BVN (Bank Verification Number) inquiry via Paystack to confirm legal identity.

`FR-PLAN-01`: Plan Assignment & Billing Logic

- **Basic (Pay-As-You-Go):** $0 monthly fee. System applies 1.5% transaction surcharge ($1.00 min, $20.00 max cap) at checkout.
- **Verapay Plus ($9.99/mo):**
  - Recurring monthly subscription billed via PayPal Subscriptions API.
  - System tracks monthly cumulative volume: 0% transaction fee up to $1,000/month; 1.5% fee applied on incremental spend.
  - Unlocks Family Group workspace, automated recurring bills, Shared Pots, and <4-hour human triage on Amber alerts.
- **Verapay Projects (2% per stage):**
  - Project creation is free; escrow funding includes a 2% verification and orchestration fee billed *strictly upon milestone sign-off*.
  - If a milestone is rejected and refunded, the 2% fee is completely waived/refunded.

4.2. Escrow & Money Movement Engine (FR-ESCROW)

`FR-ESCROW-01`: PayPal Inbound Escrow Funding

- Senders initiate payment via PayPal Checkout.
- Funds are captured into a secure merchant escrow holding balance or held as an authorization (configurable via `ESCROW_HOLD_MODE = CAPTURE_AND_HOLD`).
- Transaction transitions to `ESCROW_HELD`. Funds cannot be touched by recipient or sender until Gate 4 resolution.

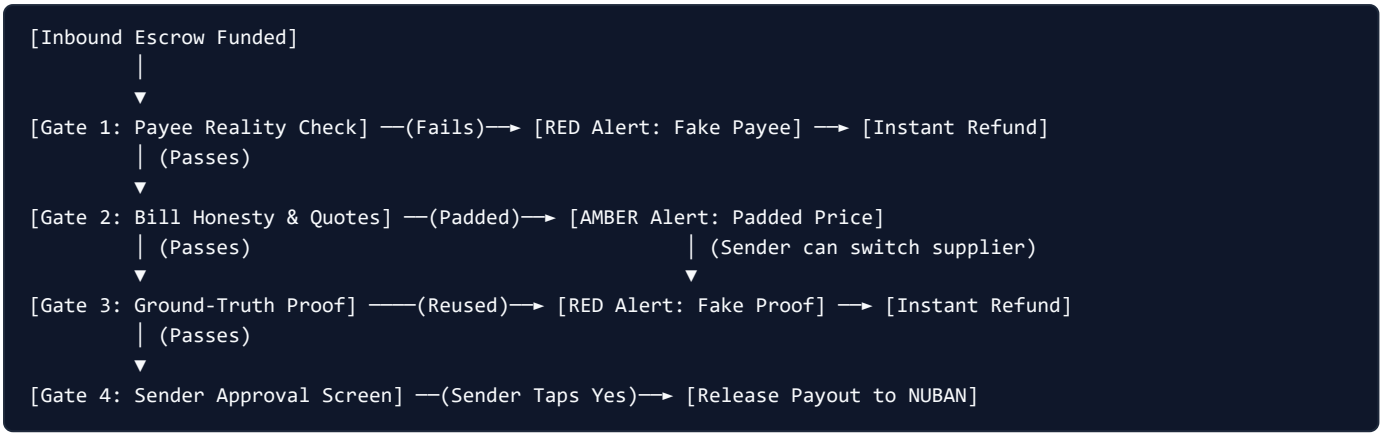
`FR-ESCROW-02`: Naira Payout Execution

- Upon sender approval (`ACTION_APPROVE`), the system triggers Paystack/Flutterwave Transfer API:
  - Generates Transfer Recipient using verified NUBAN account number and bank code.
  - Initiates Naira transfer at live guaranteed FX rate.
  - Webhook listener monitors `transfer.success`. Upon confirmation, transaction status transitions to `COMPLETED`.

`FR-ESCROW-03`: 1-Click Automated Refund

- If the sender taps `ACTION_REFUND` (or verification triggers automated red-fraud cancellation):
  - System calls PayPal Refund API (`v2/payments/captures/{capture_id}/refund`).
  - 100% of escrowed principal is credited back to sender's PayPal account/card.
  - Transaction status transitions to `REFUNDED`.

4.3. The 4-Step Verification Engine (FR-VERIFY)



`FR-VER-01`: Gate 1 – Payee Legitimacy & Registry Check

- **Corporate Registry Check:** Query Corporate Affairs Commission (CAC) API using company registration number or registered business name.
- **NUBAN Name Match:** Execute NIP account name resolution via Paystack (`GET /bank/resolve?account_number=...&bank_code=...`).
- **Match Rule:** Levenshtein similarity between CAC entity name and NUBAN bank account holder name must exceed 85%. If mismatch detected, flag `RED_PAYEE_MISMATCH`.
- **Unregistered Small Merchants:** For informal market vendors (sand, timber, blockmakers):
  - Verify seller's personal NUBAN account name.
  - Record verified merchant phone number and physical shop coordinates.
  - Enforce **Strict Pay-on-Delivery Rule**: Funds remain locked until on-site delivery is confirmed by family/foreman.

`FR-VER-02`: Gate 2 – Bill Honesty & Document OCR

- Multimodal LLM (Gemini 1.5/2.0 Flash) extracts structured JSON:
  - Student / Patient Name, Institution Name, Invoice Number, Issue Date, Due Date.
  - Line items: Description, Quantity, Unit Rate, Total Amount.
- **Duplicate Detection:** Calculates perceptual hash (dHash) and SHA-256 text fingerprint. If invoice hash matches existing record in `invoice_archive`, flag `RED_DUPLICATE_INVOICE`.
- **Arithmetic Validation:** System computes `SUM(line_items) == subtotal` and `subtotal + tax == total`. Mismatches trigger `AMBER_ARITHMETIC_ERROR`.

`FR-VER-03`: Gate 2B – Autonomous Quote Agent (Anti-Padding)

- Triggered on construction materials (cement, granite, blocks, rebar, roofing sheets).
- System extracts material specifications and location (e.g., *500 bags Dangote 42.5R Cement, Ibadan*).
- Quote Agent queries cached regional commodity index and 2–3 nearby merchant price feeds.
- **Threshold Logic:**
  - If contractor quote <= 108% of market average: `PRICE_FAIR` (Green).
  - If contractor quote > 108% and <= 125%: `PRICE_HIGH` (Amber: recommends alternative local suppliers).
  - If contractor quote > 125%: `PRICE_EXTORTION` (Amber/Red: requires contractor justification).

`FR-VER-04`: Gate 3 – Ground-Truth Execution Proof

- **Category A: Tuition, Medical, Rent:** Direct payment to CAC-verified institution; system polls for official institutional e-receipt or confirmation letter and attaches to transaction record.
- **Category B: Consumer Goods (Appliances, Provisions):** Escrow is held until the local recipient taps `CONFIRM_DELIVERY` inside the mobile app. Geolocation must be within 5km of delivery address.
- **Category C: Construction & Capital Milestones:**
  1. **In-App Camera Capture:** Media must be recorded directly via WebRTC viewfinder. Standard `<input type="file">` file uploads are disabled.
  2. **Metadata Tamper Check:** Device GPS, altitude, compass bearing, and network timestamp embedded into image EXIF and cryptographically signed.

3. **Visual Perceptual Hashing:** Checks image against database of historical project photos to prevent submitting last month's photos.
4. **Surprise 15-Second Video Prompt:** Random challenge issued: "*Record a continuous 15-second pan of the north wall showing code #4829 on a piece of paper.*"
5. **Surveyor Audit Escalation:** For stages > $2,000, sender can dispatch a certified Verapay surveyor for physical sign-off.

`FR-VER-05`: Gate 4 – Sender Approval Interface (Traffic Light UI)

- System synthesizes all signals into a clean mobile card:
  - 🟢 **GREEN:** Payee CAC confirmed, invoice price within 5% of benchmark, proof verified. Single button: `[ Approve & Pay $XXX ]`.
  - 🟡 **AMBER:** Discrepancy detected (e.g., cement quoted 18% above market, or blurry proof). Options: `[ View Cheaper Suppliers ]`, `[ Request Human Review (Plus) ]`, `[ Override & Pay ]`.
  - 🔴 **RED:** Proof failed (reused image, name mismatch, faked bill). Escrow remains locked. Option: `[ 1-Click Instant Refund ]`.

4.4. Subscription & Collaboration Features (FR-PLUS / FR-PROJ)

`FR-PLUS-01`: Automated Recurring Payments

- Senders configure monthly schedule (e.g., Mom's allowance on 1st of month, Dad's medication on 15th).
- 48 hours prior to execution, Verapay prompts provider/family for updated bill or confirms fixed schedule.
- Verapay runs Gate 1 and Gate 2 checks. If Green, sender receives a notification: "*November allowance verified. Executing in 12 hours unless cancelled.*"

`FR-PLUS-02`: Family Group Workspace

- Sender creates a `FamilyGroup` (e.g., "*The Balogun Family*").
- Invites siblings abroad (co-funders) and family members in Nigeria (beneficiaries).
- Shared real-time ledger displays all past and pending payments, itemized receipts, and verification reports.

`FR-PLUS-03`: Shared Pot (Multi-Sibling Pooling)

- Senders initiate a Shared Pot (e.g., "*Dad's Eye Surgery - $1,200*").
- Multiple siblings contribute varying amounts ($400, $500, $300) via their respective PayPal accounts.
- Funds are held in aggregate escrow until the 100% threshold is reached.
- Upon full funding, Verapay executes Gate 1–3 verification and pays the hospital directly.

`FR-PROJ-01`: Stage-by-Stage Construction Sequencing

- Projects follow standardized or custom milestone paths:
  1. *Stage 1: Site Clearing & Foundation ($2,500)*
  2. *Stage 2: Blockwork to Lintel Level ($3,200)*
  3. *Stage 3: Roof Truss & Covering ($4,000)*
  4. *Stage 4: Plastering, Electrical & Plumbing ($3,000)*
  5. *Stage 5: Finishing, Tiling & Painting ($2,800)*
- Escrow is funded stage-by-stage. Stage N+1 cannot be funded or started until Stage N proof is approved.
- Senders receive an automated weekly progress briefing written in plain English.

---

5. Non-Functional Requirements (NFR)

ID	Category	Requirement Specification
NFR-SEC-01	**Data Security**	All data in transit encrypted via TLS 1.3. Sensitive NUBAN and identity fields encrypted at rest using AES-256.
NFR-PERF-01	**Verification Latency**	Automated OCR and CAC verification must complete in `< 45 seconds`. End-to-end sender approval card ready in `< 3 minutes`.
NFR-REL-01	**Escrow Idempotency**	All payout and refund API calls must enforce UUID-based idempotency keys to guarantee zero double-disbursements.

ID	Category	Requirement Specification
NFR-NET-01	**Nigerian Network Resilience**	Mobile web application must operate on high-latency 3G networks (down to 256 kbps). In-app photo uploads must implement client-side image compression (JPEG quality 75%, max dimension 1920px) before transit.
NFR-AUD-01	**Audit Trail**	Every verification signal, EXIF coordinate, OCR token, and sender click logged to an immutable append-only audit ledger for compliance.

6. Verification Traceability Matrix

Business Goal	Functional Requirement	Test Acceptance Criteria
Eliminate Diversion	`FR-VER-01`, `FR-ESCROW-02`	100% of disbursements route to verified NUBAN account matching registered provider name.
Stop Price Padding	`FR-VER-03`	Padded cement quotes >10% over local index reliably trigger Amber warning with alternative supplier recommendations.
Catch Reused Photos	`FR-VER-04`, `FR-VIS-01`	Re-submitting an existing project image triggers Red alert with perceptual hash match score > 92%.
Guarantee Zero Fund Loss	`FR-ESCROW-03`	Senders can trigger an immediate 100% refund on any rejected or unverified transaction.